

Interacting Multipoles of Rank 3 and 4:

[illegible]

Simplify deltas

[illegible]

Simplify R

[illegible]

Simplify Multipoles

5

Exploit Traceless Conditions

6

Reduce Indices

7

Add up

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega_{\alpha\beta\gamma}\Phi_{\delta\epsilon\zeta\eta} = R_z^{-15} \cdot \frac{1}{10395} \cdot \left[\begin{array}{l} +41580 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{zzz\gamma} \\ +41580 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{zzz\beta} \\ +41580 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} \\ -1890 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ -3780 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ -5670 \cdot R_z^7 \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\delta\delta} \\ -1890 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ -3780 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ -5670 \cdot R_z^7 \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\delta\delta} \\ -11340 \cdot R_z^7 \cdot \Omega_{z\beta\gamma} \cdot \Phi_{zz\beta\gamma} \\ -1890 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} \\ -3780 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ -5670 \cdot R_z^7 \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ -11340 \cdot R_z^7 \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{zz\alpha\gamma} \\ -11340 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ -3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{zzz\gamma} \\ -3780 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{zzz\beta} \\ -3780 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{zzz\alpha} \\ +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\ +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\ +1260 \cdot R_z^7 \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \\ +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\delta\delta} \\ +1260 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +1260 \cdot R_z^7 \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\gamma\gamma} \\ +2520 \cdot R_z^7 \cdot \Omega_{\alpha\beta\delta} \cdot \Phi_{z\alpha\beta\delta} \\ +630 \cdot R_z^7 \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \end{array} \right] \quad (7)$$

Combining everything

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta}\Omega_{\alpha\beta\gamma}\Phi_{\delta\epsilon\zeta\eta} = \left[\begin{array}{l} +4 \cdot R_z^{-8} \cdot \Omega_{zz\gamma} \cdot \Phi_{zzz\gamma} \\ +4 \cdot R_z^{-8} \cdot \Omega_{zz\beta} \cdot \Phi_{zzz\beta} \\ +4 \cdot R_z^{-8} \cdot \Omega_{zz\alpha} \cdot \Phi_{zzz\alpha} \\ -2/11 \cdot R_z^{-8} \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ -4/11 \cdot R_z^{-8} \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ -6/11 \cdot R_z^{-8} \cdot \Omega_{zz\gamma} \cdot \Phi_{z\gamma\delta\delta} \\ -2/11 \cdot R_z^{-8} \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ -4/11 \cdot R_z^{-8} \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ -6/11 \cdot R_z^{-8} \cdot \Omega_{zz\beta} \cdot \Phi_{z\beta\delta\delta} \\ -12/11 \cdot R_z^{-8} \cdot \Omega_{z\beta\gamma} \cdot \Phi_{zz\beta\gamma} \\ -2/11 \cdot R_z^{-8} \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} \\ -4/11 \cdot R_z^{-8} \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ -6/11 \cdot R_z^{-8} \cdot \Omega_{zz\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ -12/11 \cdot R_z^{-8} \cdot \Omega_{z\alpha\gamma} \cdot \Phi_{zz\alpha\gamma} \\ -12/11 \cdot R_z^{-8} \cdot \Omega_{z\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ -4/11 \cdot R_z^{-8} \cdot \Omega_{\alpha\alpha\gamma} \cdot \Phi_{zzz\gamma} \\ -4/11 \cdot R_z^{-8} \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{zzz\beta} \\ -4/11 \cdot R_z^{-8} \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{zzz\alpha} \\ +4/33 \cdot R_z^{-8} \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\delta\delta} \\ +4/33 \cdot R_z^{-8} \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\ +4/33 \cdot R_z^{-8} \cdot \Omega_{z\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\gamma} \\ +2/33 \cdot R_z^{-8} \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\delta\delta} \\ +4/33 \cdot R_z^{-8} \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ +4/33 \cdot R_z^{-8} \cdot \Omega_{\alpha\beta\beta} \cdot \Phi_{z\alpha\gamma\gamma} \\ +8/33 \cdot R_z^{-8} \cdot \Omega_{\alpha\beta\delta} \cdot \Phi_{z\alpha\beta\delta} \\ +2/33 \cdot R_z^{-8} \cdot \Omega_{\alpha\alpha\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \end{array} \right] \quad (8)$$